

Question: 1

Pick any one app you use daily (Instagram, Zomato, Flipkart, etc.) and imagine you are hired to design its college campus WiFi network — write a requirements document listing at least 8 must-have features or needs from the perspective of students using that app on campus?

Answer: • 1. High Bandwidth & Throughput (Media Loading):

- Instagram par lagatar high-resolution photos, Reels aur videos load hoti hain. Network me itna bandwidth hona chahiye ki video playback bina kisi buffering ke smooth chale.

- 2. Fast Uplink Speed (Content Creation):

- Students sirf content dekhte nahi hain balki bade size ki HD Stories aur Reels upload bhi karte hain. Isliye network ki download speed ke sath-sath upload (uplink) speed bhi bahut tez honi chahiye.

- 3. Low Latency for Live Streaming (Instagram Live):

- Students campus se live sessions aur video calls karte hain. Lag aur video freeze se bachne ke liye network me latency (ping) bahut kam honi chahiye.

- 4. High-Density Access Points (APs) in Social Hubs:

- Cafeteria, college lawns, library aur auditoriums aisi jagah hain jahan sabse zyada students ek sath Instagram use karte hain. In social zones me zyada capacity wale Access Points lagane zaroori hain taaki network crash na ho.

- 5. Seamless Roaming & Handover:

- Jab students ek classroom se doosri classroom ya hostel ki taraf chalte hue Reels dekh rahe hon, toh bina connection drop hue unka Wi-Fi automatic agle nearest router par switch ho jana chahiye.

- 6. Quality of Service (QoS) Optimization:

- Wi-Fi routers me multimedia traffic (video streaming) ko traffic management ke zariye prioritize kiya jana chahiye, taaki heavy usage ke waqt bhi video delivery drop na ho.
- 7. Easy & Persistent Authentication (One-time Login):
 - Students ke liye portal login baar-baar refresh nahi hona chahiye. Ek baar authentication hone ke baad, jab tak student campus me hai, uska Instagram background me notification access ke liye connected rehna chahiye.
- 8. Content Delivery Network (CDN) / Local Caching:
 - Agar campus me koi Instagram trend ya video viral ho rahi hai aur hazaron students use ek sath dekh rahe hain, toh local network caching system hona chahiye taaki baar-baar external internet bandwidth consume na ho aur content turant load ho.

Question: 2

Sketch a preliminary network topology for a hostel building that supports uninterrupted YouTube streaming and WhatsApp video calls for 200 students — include routers, switches, access points, and label each component?

Answer: • 1. Core Router (Gateway):

- Role: Yeh hostel network ka main entry point hai jo Internet Service Provider (ISP) se high-speed fiber connection leta hai.
- Features: Isme NAT (Network Address Translation) aur heavy traffic handles ke liye dynamic routing capabilities hoti hain.

• 2. Core Switch (Layer 3 Switch):

- Role: Router se connected yeh central backbone switch hai jo pure building ke internal network traffic ko direct karta hai.

- Function: Yeh alag-alag floors ke network segmentations aur local inter-VLAN routing ko efficiently manage karta hai taaki main router par load na bade.
- 3. Floor Switches (Layer 2 Switches):
 - Role: Hostel ke har floor par ek ya do Layer 2 switches lagaye jayenge jo directly main Core Switch se Gigabit Ethernet backbone wires ke zariye jude honge.
 - Function: Yeh un floors par mojud saare Wireless Access Points (APs) ko connect karne aur PoE (Power over Ethernet) ke zariye unhe power supply dene ke kaam aate hain.
- 4. Wireless Access Points (APs):
 - Role: Har floor ke corridors aur rooms ke paas high-density Wi-Fi 6 Access Points install kiye jayenge taaki students wireless connection bana sakein.
 - Optimization: 200 students ke liye lagbhag 8 se 10 APs lagaye jayenge (har AP par ~20-25 users ka load) taaki YouTube aur WhatsApp video bina kisi drop ke smoothly chal sake.
- 5. Interconnection (Media & Bandwidth Allocations):
 - Cabling: Switches aur Access Points ke beech connections ke liye Cat6/Cat6A cables ka use kiya jayega jo 1 Gbps tak ki local speed ensure karti hain, jisse seamless video buffer possible ho pata hai.

Question: 3

Research and list 3 different types of WiFi routers available in India under ₹4000, comparing their key specs (speed, range, number of antennas, and special features like parental controls or MU-MIMO)?

1. Answer: TP-Link Archer AX12 (Wi-Fi 6) • Speed: Up to 1.5 Gbps (1201 Mbps on 5 GHz band + 300 Mbps on 2.4 GHz band). • Range: Medium to Large Home (2-3 Bedroom apartments). • Number of Antennas: 4 External High-Gain Antennas (Beamforming technology ke sath, jo devices ki taraf signal

focus karti hain). • Special Features: • Latest Wi-Fi 6 (802.11ax) technology support. • MU-MIMO & OFDMA (ek sath kayi devices par bina speed drop ke data transfer). • Advanced WPA3 Security encryption. • Custom Parental Controls (URL filtering via Tether App).

2. D-Link R15 EAGLE PRO AI (Wi-Fi 6) • Speed: Up to 1.5 Gbps (1200 Mbps on 5 GHz band + 300 Mbps on 2.4 GHz band). • Range: Medium to Large Home (AI Mesh expansion support ke sath). • Number of Antennas: 4 Fixed Omni-Directional Antennas. • Special Features: • AI Traffic & Wi-Fi Optimizer (automatically background me best channel aur bandwidth allocate karta hai). • MU-MIMO aur Smart QoS support. • Profile-based AI Parental Controls (schedules aur website blocking options). • Voice Control integration (Works with Amazon Alexa & Google Assistant).
3. TP-Link Archer C80 (Wi-Fi 5 AC1900) • Speed: Up to 1.9 Gbps (1300 Mbps on 5 GHz band + 600 Mbps on 2.4 GHz band) — Legacy Wi-Fi 5 me iski peak speed zyada hai. • Range: Medium Home (Highly optimized dual-band coverage). • Number of Antennas: 4 External High-Performance Antennas. • Special Features: • 3x3 MU-MIMO Technology (router 3 alag devices ke sath ek hi samay par concurrent stream handle kar sakta hai). • Airtime Fairness (purane devices ki wajah se network ko slow hone se rokta hai). • Full Gigabit Ports (high-speed wired PC/Console gaming ke liye). • Comprehensive Parental Controls (Time management aur URL controls tether app ke zariye).

Question: 4

Meet with a 'client' (ask a friend or family member to act as a student user) and conduct a 10-minute interview to gather their top 5 pain points with current campus internet or mobile data usage. Summarize their responses and suggest at least 2 design features that would address their concerns?

Answer: • 1. High Congestion during Peak Hours:

• Lunch breaks aur evening times me jab sabhi students ek sath connect hote hain, toh internet ki speed bilkul drop ho jaati hai.

- 2. Frequent Disconnections while Moving (Roaming Issues):
 - Hostel se campus buildings ya library ki taraf walk karte waqt Wi-Fi signals lagatar disconnect hote hain aur login portal baar-baar samne aa jata hai.
- 3. Dead Zones in Indoor Corners:
 - Purane stone structures aur bade lecture halls ke kone wale rooms me network connectivity bilkul nahi milti (zero coverage).
- 4. Heavy Latency during Online Tests/Submissions:
 - College portal par quiz ya assignment submit karte waqt heavy lag ke kaaran session expiry ya failed submission ka darr bana rehta hai.
- 5. Unsecure Open Wi-Fi Exposure:
 - Campus ka open Wi-Fi network use karne par security aur personal banking details leak hone ka lagatar khatra rehta hai.

Question: 5

Combine your requirements document and topology sketch into a single PDF file and submit it to your mentor or instructor for feedback.

Hint: Make sure your document is clear, organized, and uses simple diagrams or tables where possible.?

Answer: • Document Structure Checklist:

- Section 1: App Specifications (Instagram) - Isme video streaming aur dynamic updates ke liye bandwidth aur low-latency specs shamil hain (Question 1 ka reference).
- Section 2: Network Infrastructure Blueprint - Isme Core Router, Layer 3 Switches, aur Floor Access Points (Wi-Fi 6) ka multi-tier structural details aur sketch mapped hai (Question 2 ka reference).

- Section 3: Budget Hardware Analysis - India me under ₹4000 aane wale top routers ka technical data analysis design sheet me integration (Question 3 ka reference).
- Section 4: User Experience Metrics - 10-minute client feedback key points aur unke core design implementations embedded hain (Question 4 ka reference)